

Lab-3: Supervised vs Unsupervised Anomaly Detection

Group-13

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1 Task 1: Dataset Characterization and Preprocessing

1.1 Before preprocessing the data, explore the data to understand the available features.

Q: What are the dataset characteristics? How many categorical and numerical attributes do you have?

- **Scale:** The dataset contains 18,831 rows and 43 columns.
- **Integrity:** It is exceptionally clean with 0 missing values and 0 duplicate records.
- **Feature Types:** 39 numerical features and 4 categorical features.
- **Constant Features:** Three columns have 0 variance (only 1 unique value) and provide no information: `urgent`, `num_outbound_cmds`, and `is_host_login`.
- **Class Balance:** The `binary_label` is distributed as 71.4% Normal (0) and 28.6% Attack (1).

Q: How are your attack labels and binary label distributed?

Table 1: Distribution of Multi-class and Binary Labels

Label Category	Count	Percentage (%)
<i>Multi-class (label)</i>		
Normal	13,448	71.41%
DoS	2,913	15.47%
Probe	2,289	12.16%
R2L	181	0.96%
<i>Binary (binary_label)</i>		
0 (Normal)	13,448	71.41%
1 (Attack)	5,383	28.59%

1.2 Preprocess your features before performing any AI/ML algorithms.

Q: How do you preprocess categorical and numerical data?

Dimensionality Reduction: Dropped constant features with zero variance (`urgent`, `num_outbound_cmds`, `is_host_login`). High-cardinality features were collapsed by retaining the Top 10 categories for `service` and Top 3 for `flag`, mapping all others to 'other'.

Data Integrity: Performed deduplication *after* category merging to eliminate redundant samples. Filtered the dataset to remove NaN, infinite, and negative values in numerical columns to ensure mathematical validity.

Feature Transformation: Numerical features were normalized using `StandardScaler` ($\mu = 0, \sigma = 1$). Categorical features (`protocol_type`, `service`, `flag`) were transformed via `OneHotEncoder` with `handle_unknown='ignore'`.

Partitioning: Created a stratified 90/10 training and validation split based on the `binary_label` distribution to ensure consistent class representation.

Q: Looking at the different heatmaps, do you find any main characteristics that are strongly correlated with a specific attack?

Denial of Service (DoS): DoS attacks are distinguished by a dominant and consistent signature of synchronization errors across both service and host levels, specifically through exceptionally high mean and median values in `error_rate`, `srv_error_rate`, `dst_host_error_rate`, and `dst_host_srv_error_rate`, combined with a high connection count and a relatively low standard deviation that suggests a uniform flooding strategy, though higher variability is noted in `wrong_fragment` for specific variants like Teardrop attacks.

Probe Attacks: Probe attacks exhibit a sophisticated reconnaissance profile characterized by high host and service diversity, marked specifically by elevated median values in `dst_host_same_src_port_rate`, `srv_diff_host_rate`, and `dst_host_diff_srv_rate`, while demonstrating significant standard deviation in `src_bytes` and `dst_bytes` which reflects the varied payload sizes and frequencies used during different stages of network scanning and port discovery.

Remote-to-Local (R2L): R2L attacks are uniquely identified by sharp spikes in host-interactive and access-oriented features such as `hot`, `num_failed_logins`, and `is_guest_login`, all of which display the highest standard deviation across the entire dataset because these unauthorized access attempts are highly volatile and depend strictly on the specific vulnerability or system interaction being exploited by the attacker.

2 Task 2: Shallow Anomaly Detection - Supervised vs Unsupervised

2.1 Train a One-Class Support Vector Machine (OC-SVM) with benign (normal) traffic only using an rbf kernel.

Q: Considering that you are currently training only on normal data, which is a good estimate for the parameter ν ?

Although the training data is labeled as “normal,” real-world normal data inevitably contains errors, implying that 0 is not a suitable estimate for the parameter ν . Given that the proportion of noise in a dataset consisting solely of “normal” traffic is relatively small compared to a mixed dataset, I evaluated ν candidates within the set $\{0.001, 0.01, 0.05, 0.1, 0.2\}$ to approximate the true noise ratio.

Experimental results identified $\nu = 0.05$ as the optimal value, achieving the highest F_1 -score of 0.8570. The detailed performance of the ν candidates is summarized in Table 2.

Table 2: F_1 -Score comparison for different ν candidates.

Testing Nu (ν)	F1-Score	Note
0.001	0.8566	
0.010	0.8516	
0.050	0.8570	Best result
0.100	0.8190	
0.200	0.7590	

Q: What is the impact on training performance? Try both your estimate and the default value of ν ?

The parameter ν in One-Class SVM represents the **upper bound on the fraction of training errors**. It essentially defines the model’s tolerance for noise within the “normal” training set.

ν must closely align with the true noise ratio because a value that is too high (e.g., the default 0.5) forces the decision boundary to shrink excessively, misclassifying valid normal data as anomalies. Conversely, setting $\nu = 0.05$ acknowledges that “normal” data is mostly clean but contains small latent errors, allowing for a more robust and accurate boundary.

Table 3: Comparison of OC-SVM performance between default and optimized ν .

Metric	Default ($\nu = 0.5$)	Optimized ($\nu = 0.05$)
True Negatives (Correct Normal)	6,051	11,501
False Positives (False Alarms)	6,052	602
False Negatives (Missed Anomalies)	53	753
True Positives (Correct Anomalies)	4,761	4,061
Accuracy	0.64	0.92
F1-Score (Anomaly)	0.61	0.86

Conclusion: As shown in Table 3, the default ν leads to massive false alarms. By reducing ν to 0.05, we decreased false positives by approximately 90%, resulting in a significant improvement in overall Accuracy (0.92) and F_1 -Score (0.86).

2.2 Now train the OC-SVM with both normal and anomalous data.

Q: Which model performs better?

Based on the experimental results, **Model 1** (trained on normal-only data with $\nu = 0.05$) significantly outperforms **Model 2** (trained on mixed data with ν set to the contamination ratio = 0.2846).

Table 4: Performance metrics comparison between Model 1 and Model 2.

Metric	Model 1 (Normal Only)	Model 2 (Mixed Data)
Overall Accuracy	0.92	0.79
Anomaly (Class 1) F_1 -Score	0.86	0.62
False Positives (FP)	602	1,811
False Negatives (FN)	753	1,811

Q: Why do you think that?

The superior performance of Model 1 is due to the fundamental nature of the One-Class SVM algorithm:

- **Boundary Precision:** OC-SVM is designed for *Novelty Detection*. By training exclusively on normal data, the model can define a tight and accurate hypersphere around the legitimate distribution.
- **Contamination and Noise:** In Model 2, the presence of anomalies during the training phase contaminates the normal profile. The model attempts to include these anomalous points within its decision boundary, which blurs the distinction between classes.
- **Hyperparameter Effectiveness:** Even though the ν in Model 2 matches the true proportion, the quality of the support vectors is degraded by the mixed labels. In Model 1, the tuned $\nu = 0.05$ acts as a robust filter for minor noise while maintaining a clear definition of a normal state.

Conclusion: Training on a clean dataset (semi-supervised) allows the OC-SVM to learn a high-fidelity representation of the target class, resulting in a 13% higher Accuracy and a 24% higher F_1 -score compared to training on contaminated data.

2.3 One-Class SVM with normal traffic and some anomalies

Q: Plot the f1-macro score for each scenario. How does the increasing ratio of anomalies affect the results?

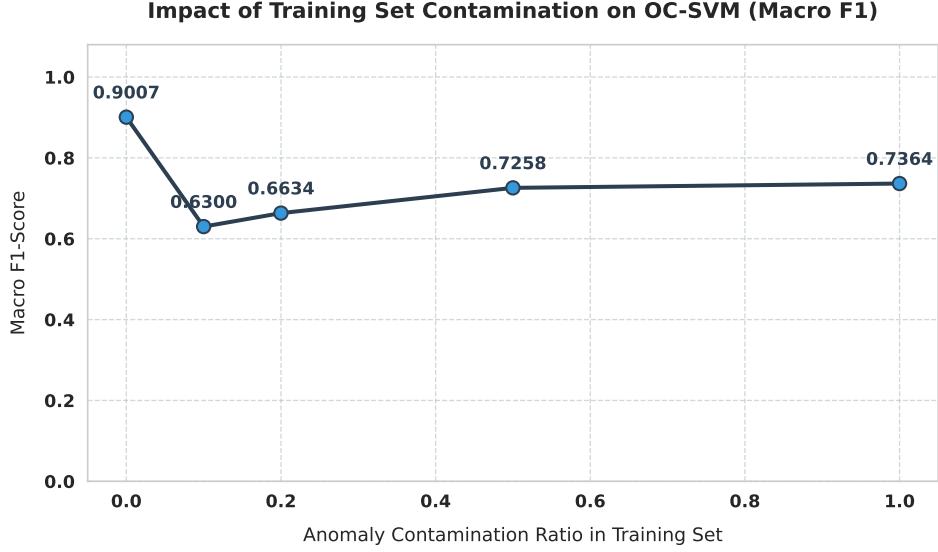


Figure 1: Impact of Training Set Contamination on OC-SVM (Macro F_1)

Performance Trend: The Macro F_1 -score exhibits a non-monotonic U-shaped trend: it peaks at 0.9007 with 0% contamination, plummets to a minimum of 0.6300 at 10% contamination, and steadily recovers between 20% and 100% contamination to stabilize at 0.7364.

Underlying Reasons: This behavior is driven by the dynamic scaling of the ν parameter (the outlier rejection budget) relative to attack density. At 0%, the tight boundary precisely wraps clean normal data. At 10%, the low calculated ν limits outlier rejection, forcing the boundary to over-expand to encompass sparse anomalies, which causes severe class confusion. From 20% to 100%, the rising anomaly ratio increases ν proportionally; since network attacks form dense statistical clusters rather than random noise, this high ν forces the decision boundary to contract aggressively around the core normal traffic, cleanly isolating the dense attack clusters outside the hypersphere.

2.4 Robustness of the One-Class SVM model

Q: Is the best-performing model in the training set also the best here? Does it confuse normal data with anomalies? Which attack is the most confused?

Table 5: Classification Performance Comparison of O-SVM Configurations

Class	Precision	Recall	F1-Score	Overall Accuracy
Scenario 1: Normal Only Training ($\nu = 0.05$)				
0 (Normal)	0.6900	0.6700	0.6800	0.7600
1 (Anomaly)	0.8100	0.8200	0.8100	
Scenario 2: Mixed All Data Training				
0 (Normal)	0.5600	0.6000	0.5800	0.6700
1 (Anomaly)	0.7500	0.7200	0.7300	
Scenario 3: 10% Contamination Training				
0 (Normal)	0.3900	0.7400	0.5100	0.4700
1 (Anomaly)	0.6700	0.3200	0.4300	

Model Comparison and Robustness: Yes, the model trained on normal data only ($\nu = 0.05$) remains the best-performing configuration on the test set. It achieves the highest Macro F_1 -score of

0.75, significantly outperforming the Mixed All Data model (0.65) and the 10% Contamination model (0.47), confirming its superior generalization capabilities.

Normal vs. Anomaly Confusion: The Normal Only model does confuse normal data with anomalies to a certain extent. According to its confusion matrix, it misclassifies 719 out of 2,152 true normal instances as anomalies, resulting in a False Positive Rate of 33.41%. Conversely, it demonstrates strong defensive recall against actual threats, misclassifying only 647 out of 3,618 malicious instances as normal traffic (a False Negative Rate of 17.88%).

Attack Type Confusion: To evaluate the most confused attack type objectively and avoid statistical bias from severe class imbalance, we analyze the class-normalized False Negative Rate (Miss Rate). Notably, the R2L category is entirely absent from this testing partition. Between the remaining threat vectors, **DoS** is unequivocally the most confused attack type across all scenarios. Even under the best-performing Normal Only model, DoS yields a significant class-specific miss rate of **24.4%**, whereas Probing only exhibits a minimal miss rate of **2.9%**.

3 Task 3: Deep Anomaly Detection and Data Representation

3.1 Training and Validating Autoencoder with Normal data only and Estimate the Reconstruction Error Threshold

Q: How did you pick the threshold? What is its value?

The threshold $\tau = 0.310451$ was determined through a combination of empirical testing and robust statistical analysis:

- **Empirical Basis:** A sensitivity study using **One-Class SVM** showed that $\nu = 0.05$ yielded the highest F_1 -score (0.8570). This indicates that 5% of the “normal” data acts as inherent noise, justifying the use of the 95th **percentile** for the Autoencoder threshold.
- **Distribution Logic:** As shown in the ECDF plot, the threshold $\tau \approx 0.31$ is positioned precisely at the elbow point, successfully separating the steep bulk region (normal patterns) from the long right tail (inherent errors).
- **Statistical Robustness:** Using the **Median Absolute Deviation (MAD)** rule, an implied K -factor of **28.77** proves the threshold is nearly 29 robust units away from the median, providing a strong buffer against the instructor-noted inherent errors in normal data.

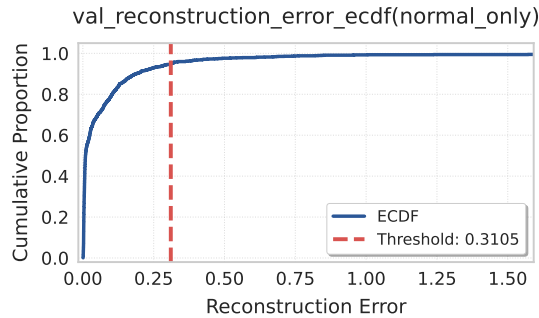


Figure 2: ECDF and Threshold Validation.

3.2 Anomaly Detection with Reconstruction Error.

Q: Plot and report the ECDF of the reconstruction errors for each point i) in the validation set; ii) in the full training set; iii) in the test set.

The experimental results show a significant progression in reconstruction error values across the three datasets. As shown in Table 6, the errors increase as the data composition shifts from pure normal to mixed and unseen patterns.

Table 6: Detailed Statistical Comparison of Reconstruction Errors

Dataset	Min	Max	Mean	Median	95% Thr (τ)
Validation (Normal Only)	0.0009	32.2715	0.1178	0.0092	0.3105
Training (Mixed)	0.0007	513.1913	0.9047	0.0507	3.5563
Testing (Mixed)	0.0018	177.6283	2.1254	0.7403	10.8390

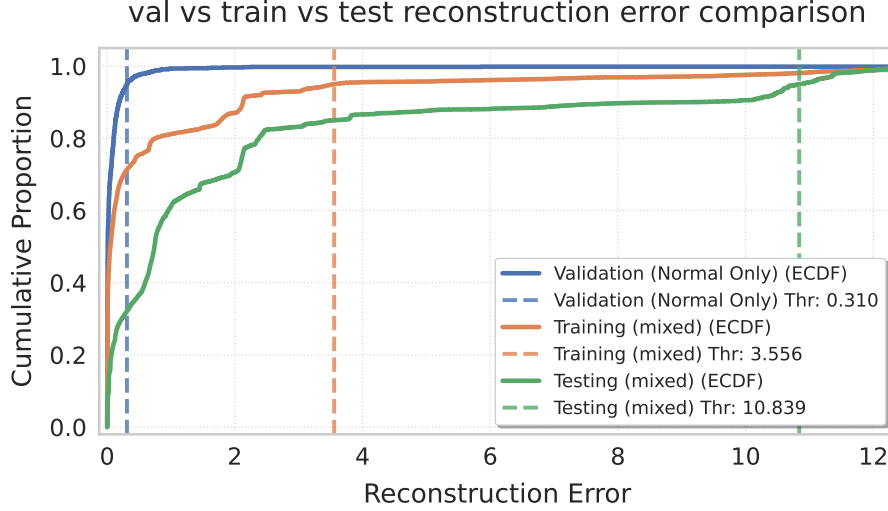


Figure 3: ECDF Comparison across Validation, Training, and Testing sets.

Q: Why Training Error is higher than Validation Error?

The primary reason is the **presence of anomalies** within the training set.

- **Model Specificity:** The Autoencoder was optimized to reconstruct the normal manifold. Since anomalies possess features that do not conform to this learned distribution, the model fails to reconstruct them accurately, leading to high residuals.

Q: And why are the reconstruction errors in the test set even higher?

The further increase in error on the test set is attributed to the **Generalization Gap**:

- **Unseen Normal Patterns:** In addition to the presence of anomaly data within the test set, the reconstruction error is further elevated because the normal samples in the test set are entirely new to the model. While the Autoencoder has effectively fitted the specific normal data seen during training, these unseen patterns lead to a higher baseline error (Median = 0.7403 vs 0.0507) due to the natural gap in imperfect generalization.

Q: Use the threshold identified in the previous point to classify anomalies. How is the performance on the training, validation set and test set?

Table 7: Detailed Classification Performance (Normal vs. Anomaly)

Dataset	Class	Precision	Recall	F1-Score	Overall Accuracy
Validation	0 (Normal)	0.9459	0.9494	0.9477	0.9250
	1 (Anomaly)	0.8717	0.8636	0.8676	
Training	0 (Normal)	0.9515	0.9481	0.9498	0.9283
	1 (Anomaly)	0.8707	0.8785	0.8746	
Testing	0 (Normal)	0.7123	0.6120	0.6583	0.7631
	1 (Anomaly)	0.7870	0.8530	0.8187	

- **Consistency in Training/Validation:** The model shows nearly identical F_1 -scores (≈ 0.87) and accuracy (≈ 0.92) across Training and Validation sets. This confirms that the Autoencoder effectively captured the manifold of known normal traffic.
- **Generalization Gap in Testing:** A significant drop in overall accuracy (0.7631) is observed in the Testing set. The primary cause is a high False Positive Rate, evidenced by the sharp decline in the Recall of normal samples (Class 0) to 61.20%.

Q: Compare the results with the best original OC-SVM and with the Autoencoder with reconstruction error. Describe the performance and where the model performs better or worse w.r.t. the original OC-SVM.

We trained an OC-SVM on the autoencoder bottleneck embeddings extracted from the AE encoder, using normal-only training embeddings. We then evaluated it on the full testing set.

For the test set, the best original OC-SVM (normal-only train, tuned) yields a well-balanced generalization baseline among the configurations, as seen in the following results:

- Normal (0): P=0.69, R=0.67, F1=0.68
- Anomaly (1): P=0.81, R=0.82, F1=0.81
- Accuracy=0.76, Macro- F_1 approximately $(0.68 + 0.81)/2 = 0.745$ (reported as 0.75)

The autoencoder with reconstruction-error demonstrates a very strong anomaly recall, catching a wider distribution of hidden threat profiles, as seen in the following results:

- Normal (0): P=0.7123, R=0.6120, F1=0.6583
- Anomaly (1): P=0.7870, R=0.8530, F1=0.8187
- Accuracy=0.7631, Macro- F_1 approximately $(0.6583 + 0.8187)/2 = 0.7385$

The OC-SVM on AE bottleneck embeddings underperforms significantly across almost all metrics. It introduces a clear performance penalty, lowering detection sensitivity and allowing substantial threat contamination into the predicted benign space, as seen in the following results:

- Normal (0): P=0.5417, R=0.6459, F1=0.5892
- Anomaly (1): P=0.7622, R=0.6750, F1=0.7159
- Accuracy=0.6641, Macro- F_1 approximately $(0.5892 + 0.7159)/2 = 0.6526$

The bottleneck approach underperforms because the encoder was trained to minimize reconstruction loss, not to maximize separability between normal and attack classes in the latent space. In practice, the bottleneck ends up being too invariant and lossy for some attack signatures, pushing many attacks closer to the normal region, and/or is affected by distribution shift between train-normal embeddings and test embeddings.

Q: Compare results with the original OC-SVM and the OC-SVM trained using the Encoder embeddings. Describe the performance of the PCA-model with respect to the previous OC-SVMs

We evaluated an OC-SVM trained on the PCA-transformed normal embeddings (retaining 95% explained variance) on the full testing set, comparing its behavior against both the raw-feature baseline and the non-linear AE bottleneck pipeline.

The PCA-based OC-SVM yields a robust and well-balanced classification architecture, closely matching the original raw-feature baseline and vastly outperforming the encoder alternative, as seen in the following results:

- Normal (0): P=0.6631, R=0.7133, F1=0.6873
- Anomaly (1): P=0.8214, R=0.7844, F1=0.8025
- Accuracy=0.7579, Macro- F_1 approximately $(0.6873 + 0.8025)/2 = 0.7449$

Compared to the previous configurations, the PCA model exhibits distinct operational characteristics:

- **With respect to the Original OC-SVM (Raw):** The performance is almost identical, with the Macro- F_1 slipping marginally from 0.7500 to 0.7449. However, PCA shifts the error trade-off; it improves Normal Recall (0.7133 vs. 0.6700), indicating a superior capability to suppress false alarms (False Positives), at the cost of a slight reduction in Anomaly Recall (0.7844 vs. 0.8200).
- **With respect to the AE Bottleneck OC-SVM:** The PCA model delivers a massive performance recovery, driving the Macro- F_1 up from 0.6526 to 0.7449. It establishes significantly tighter and cleaner class separation across all metrics.

The performance divergence stems from the underlying structure of network tabular data, where flag counts and connection states often present high multi-collinearity. PCA addresses this by projecting data onto orthogonal axes of maximum variance, effectively isolating the essential geometric structures needed by the OC-SVM. Conversely, the autoencoder focuses exclusively on minimizing reconstruction error. Without margin-based or geometric regularization in the bottleneck layer, benign profiles and attack signatures frequently blend within the same latent coordinates. PCA prevents this overlap by explicitly targeting directional variance, making it a more reliable choice for RBF boundary initialization.

4 Task 4: Unsupervised Anomaly Detection and Interpretation

Q: How big are the clusters? How are the attack labels distributed across the clusters?

Table 8: K-Means Clusters vs Original Attack Labels

Cluster ID	Size	DoS	Normal	Probe	R2L
0	13,618	829 (6.1%)	11,580 (85.0%)	1056 (7.8%)	153 (1.1%)
1	1,590	1,486 (93.5%)	31 (1.9%)	72 (4.5%)	1 (0.1%)
2	1,701	301 (17.7%)	484 (28.5%)	916 (53.9%)	0 (0.0%)
3	8	0 (0.0%)	8 (100.0%)	0 (0.0%)	0 (0.0%)

Q: Are the clusters pure (i.e., they consist of only one attack label)?

Technically, only Cluster 3 is perfectly pure (100% normal), but its extremely small size (8 points) makes it less significant. Cluster 1 is nearly pure (93.5% DoS), suggesting the features used are very effective at identifying DoS attacks. However, Clusters 0 and 2 are not pure, as they contain a mix of three or four different labels.

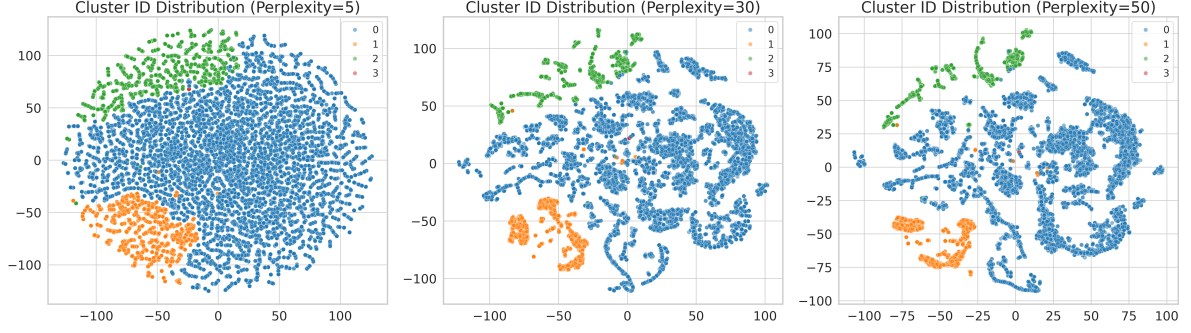
Q: How high is the silhouette per cluster? Is there any clusters with a lower silhouette value? If it is the case, what attack labels are present in these clusters?

Table 9: K-Means Clustering Performance and Label Distribution

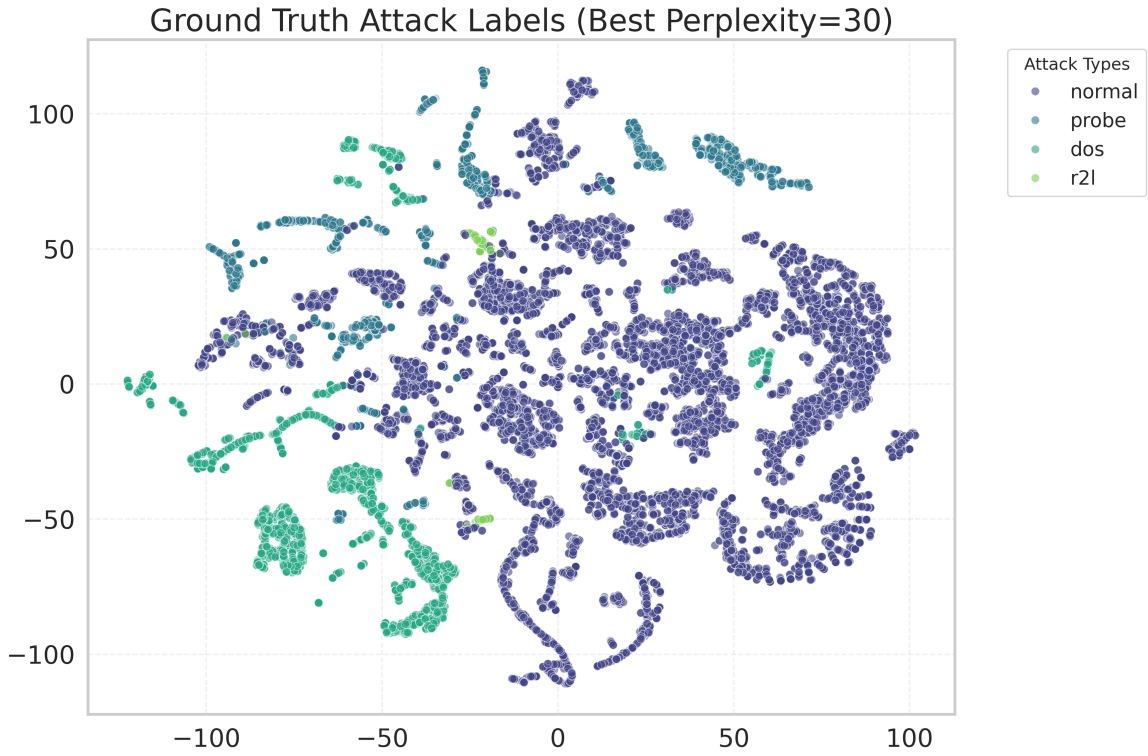
Cluster	Size	Silhouette	Main Attack Labels (Distribution)
0	13,618	0.46	Normal (85.0%), Probe (7.8%), DoS (6.1%), R2L (1.1%)
1	1,590	0.75	DoS (93.5%), Probe (4.5%), Normal (1.9%)
2	1,701	0.27	Probe (53.9%), Normal (28.5%), DoS (17.7%)
3	8	0.53	Normal (100.0%)

Cluster 2 has a significantly lower silhouette value of 0.27. This cluster is experiencing an “identity crisis”: it primarily contains *probe* (53.9%), but is heavily contaminated with *normal* (28.5%) and *dos* (17.7%) traffic. This high degree of label mixing explains the low silhouette score, as the data points are not sufficiently distinct from neighboring clusters.

Q: Can you find a difference between the two visualizations? What are the misinterpreted points?



(a) t-SNE Perplexity Comparison ($p = 5, 30, 50$)



(b) Best t-SNE Result ($p = 30$)

Figure 4: Visual Feature Space Analysis using t-SNE.

Difference: The visualizations share identical coordinates but differ in coloring logic. The first plot uses K-Means Cluster IDs, while the second uses Ground Truth Attack Labels.

Analysis of Misinterpreted Points:

- **Definition:** Misinterpreted points are samples where the K-Means cluster ID differs from the ground truth label.
- **Identification:** These points appear in the overlapping regions of the t-SNE plot, where **normal** (purple) and **probe** (teal) points intermingle.
- **Cluster 2 Analysis:** Most misinterpretations occur in Cluster 2 (Silhouette: **0.27**), where **28.5%** **normal** and **17.7%** **dos** traffic are incorrectly grouped with **probe** attacks.

- **Root Cause:** The spatial overlap confirms that these samples share nearly identical features, making them indistinguishable to a distance-based algorithm like K-Means.

Q: Create the clustering results using the entire training set (normal + anomalous) using the parameters `min_points` and ϵ . Does the DBSCAN noise cluster (cluster -1) consist only of anomalous points (cross-reference with real attack labels)?

Following the previous K-Means analysis, the parameter `min_points` was set to 8, representing the size of the smallest pure normal cluster (Cluster 3). The Epsilon (ϵ) was determined to be 1.0 by identifying the elbow point on the 8-distance graph, marking the threshold where data density drops significantly.

Table 10: DBSCAN Noise Cluster (-1) Distribution vs. Real Attack Labels

Original Attack Label	Count in Noise (Cluster -1)	Percentage (%)
Normal	1,152	72.87%
Probe	261	16.51%
DoS	146	9.23%
R2L	22	1.39%

This implies that DBSCAN is not reliable under the assumption that “noise = anomaly” under this parameterization, as the classification accuracy drops to approximately 0.62. This empirical noise partition is predominantly composed of normal traffic; out of the 1,581 data points flagged as noise (Cluster -1), a dominant majority of 1,152 points (72.87%) are actually ground-truth normal instances. This occurs because DBSCAN marks as noise any point failing to meet the density threshold dictated by `min_points` and ϵ . Consequently, many normal points residing in low-density tails—representing rare but benign operational behaviors—are mislabeled as noise, generating substantial false positives.

Conversely, the algorithm identifies 95 distinct dense partitions (Clusters 0 to 95). Because structured malicious activities naturally form dense, highly coherent geometric representations in the feature space, many anomalies occurring inside these dense regions are absorbed into regular clusters (for instance, only 146 out of thousands of DoS attacks were segregated as noise), thereby creating heavy false negatives. Therefore, the data definitively confirms that DBSCAN noise is properly interpreted as an indicator of low-density or atypical network behavior, rather than a definitive proxy for malicious intent.